

DATA PROFILER

Part A: Fundamentals — Task Answers

Fundamentals of Data Analysis, Data Science Projects & Tensors

1. Write a short note : What is Data Analysis?

Data Analysis is the process of inspecting, cleaning, transforming, and interpreting raw data with the goal of discovering useful information, drawing conclusions, and supporting decision-making. It converts unstructured or scattered data points into meaningful insights that individuals and organisations can act upon.

In simple terms, it answers questions such as “What happened?”, “Why did it happen?”, “What will happen next?”, and “What should we do about it?” by examining patterns, trends, and relationships hidden inside data.

Typical stages of Data Analysis :-

- Collecting raw data from relevant sources (databases, sensors, surveys, logs, etc.)
- Cleaning the data to remove errors, duplicates, and missing values
- Exploring and analysing the data using statistical or visual techniques
- Interpreting the results to generate actionable insights
- Communicating findings through reports, dashboards, or visualisations



Fig 1.1 — The Data Analysis process flow

2. Describe how to plan a Data Science Project , Listing all steps.

A Data Science project is executed in a structured, iterative manner rather than as a single linear task. Careful planning at each stage ensures the final model or analysis is reliable, relevant, and deployable in a real business setting. The steps below outline a complete, industry-standard plan.

Step	Description
1. Problem Definition	Clearly define the business problem, objectives, and success metrics. Understand what question needs to be answered and how the outcome will be used.

Step	Description
2. Data Collection	Gather relevant data from internal databases, APIs, third-party sources, logs, or surveys.
3. Data Cleaning & Preparation	Handle missing values, remove duplicates/outliers, correct inconsistent formats, and structure the data for analysis.
4. Exploratory Data Analysis (EDA)	Use statistics and visualisations to understand distributions, correlations, and patterns in the data.
5. Feature Engineering	Create, select, and transform variables (features) that improve model performance.
6. Model Building	Select suitable algorithms and train machine learning / statistical models on the prepared data.
7. Model Evaluation	Assess model performance using metrics such as accuracy, precision, recall, RMSE, etc., and validate on unseen data.
8. Deployment	Integrate the trained model into a production system, application, or dashboard.
9. Monitoring & Maintenance	Continuously track model performance in production and retrain/update as data or business needs change.

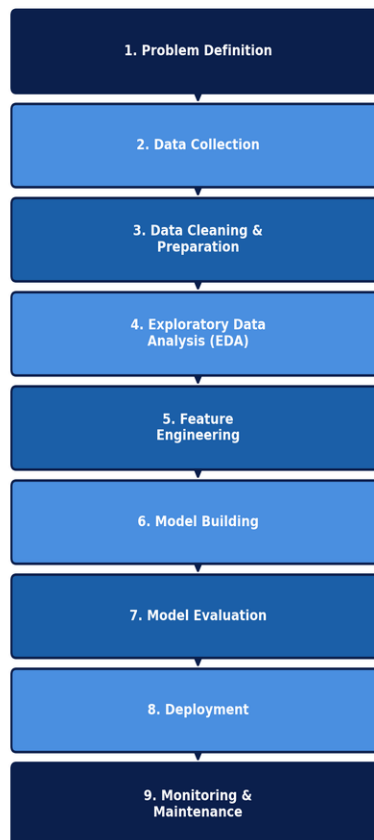


Fig 2.1 — The Data Science project lifecycle

3. Frame a Machine Learning Problem Statement : Predict whether Customer will Churn based on purchase behavior .

A well-framed ML problem statement clearly defines the objective, the type of learning task, the inputs, and the expected output before any modelling work begins.

Objective	Predict whether an existing customer is likely to churn (stop using the product/service) in the near future, based on their past purchase behaviour.
Problem Type	Supervised Learning — Binary Classification (Churn = Yes / No)
Target Variable	“Churn” — a binary label (1 = customer churns, 0 = customer stays)
Input Features	Purchase frequency, recency of last purchase, average order value, browsing/engagement activity, number of support tickets raised, customer tenure, discount usage, etc.
Output	A probability score / class label indicating churn risk for each customer
Evaluation Metrics	Accuracy, Precision, Recall, F1-score, and ROC-AUC (since churn datasets are usually imbalanced)
Business Use	Enables the company to proactively target high-risk customers with retention offers, reducing revenue loss

Visual representation

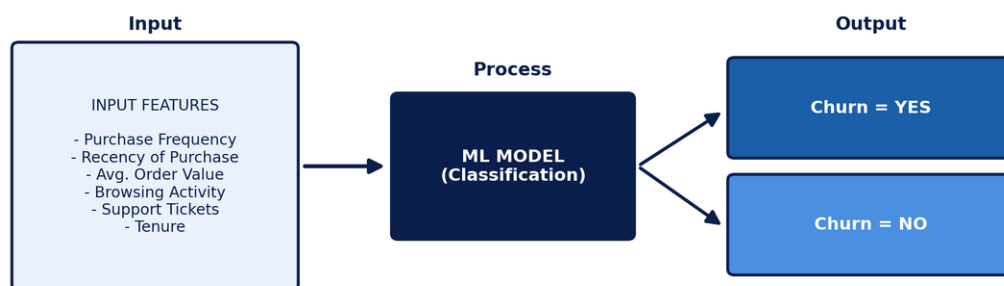


Fig 3.1 — Churn prediction as an ML input → model → output pipeline

4. Explain :-

4.1 What are Tensors?

A tensor is a generalised, multi-dimensional container for numerical data. It is the fundamental data structure used in Machine Learning and Deep Learning frameworks (such as NumPy, TensorFlow, and PyTorch) to store and manipulate data of any dimensionality.

Every tensor has a rank (also called order or number of dimensions), a shape (the size along each dimension), and a data type (such as int32 or float64). Scalars, vectors, and matrices are simply special, low-rank cases of the more general tensor concept.

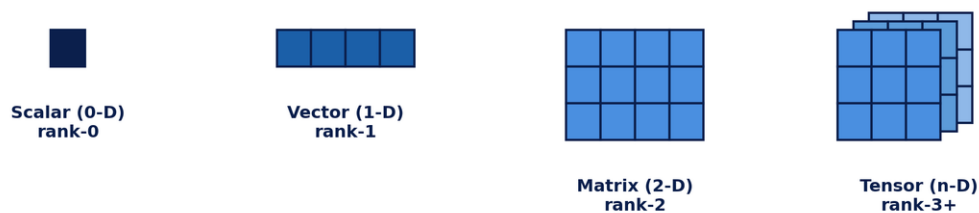


Fig 4.1 — Scalars, vectors, matrices, and tensors as increasing ranks of the same idea

4.2 Provide an In-depth explanation of each tensor with simple examples using Numpy .

a) Scalar — Rank-0 Tensor

A scalar is a single number with no axes at all — it has zero dimensions. In NumPy terms, a scalar tensor simply holds one value (for example, the number 5) and its number of dimensions (ndim) is 0, with an empty shape.

b) Vector — Rank-1 Tensor

A vector is a one-dimensional array of numbers arranged along a single axis — for example, the four values 10, 20, 30, and 40 representing feature values for one data sample. In NumPy, such a vector has `ndim = 1` and a shape describing how many elements it contains, such as `(4,)`.

c) Matrix — Rank-2 Tensor

A matrix is a two-dimensional array arranged in rows and columns — commonly used to represent a full dataset, where each row is a sample and each column is a feature. A matrix with 2 rows and 3 columns has `ndim = 2` and a shape of `(2, 3)` in NumPy.

d) Higher-rank Tensor — Rank-3 and above

A rank-3 (or higher) tensor extends this idea further by stacking matrices along additional axes. A common real-world example is a batch of colour images, where the four dimensions represent the number of images in the

batch, image height, image width, and colour channels (for instance, 32 images of 64×64 pixels in 3 colour channels would have shape (32, 64, 64, 3) and ndim = 4).

Summary table

Tensor	Rank	Example Shape	Real-world Example
Scalar	0	()	A single value, e.g. temperature = 36.6
Vector	1	(4,)	A list of 4 feature values for one customer
Matrix	2	(2, 3)	A small dataset: 2 samples × 3 features
3-D Tensor	3	(2, 2, 2)	A single 2×2 colour image with 2 channels
n-D Tensor	$n \geq 4$	(32, 64, 64, 3)	A batch of 32 RGB images, 64×64 pixels each

End of Part A — Fundamentals